1.3 Classification of Living World

From the previous chapter we know that the planet earth is endowed with living and non -living things. In our daily life we see several things in and around us. Imagine, you are on a trip to Hill station. You are enjoying the beauty of mountains, dazzling colour of the flowers, and melodious sound of the birds. You may be capturing most of the things you come across in the form of photography. Now, from this experience can you mention the objects you have come across? Can you record your observations and tabulate them?. How will you organize the things? Will you place mountain and flowers together or tall trees and trailing herbs in one category or place it in different category? If you place it in different category, what made you to place them in different category? So classification is essential and could be done only by understanding and comparing the things based on some characters. In this chapter we shall learn about classification of living world.

Many attempts have made in the past to classify the organisms on earth. **Theophrastus**, "Father of Botany" used the morphological characters to classify plants into trees, shrubs and herbs. Aristotle classified animals into two groups. i.e., *Enaima* (with red blood) and *Anaima* (without red blood).

Carl Linnaeus classified living world into two groups namely Plants and Animals based on morphological characters. His classification faced major setback because Prokaryotes and Eukaryotes were grouped together. Similarly fungi, heterotrophic organisms were placed along with the photosynthetic plants. In course of time, the development of tools compelled taxonomists to look for different areas like cytology, anatomy, embryology, molecular biology, phylogeny etc., for classifying organisms on earth. Thus, new dimensions to classifications were put forth from time to time.

1.3.1 Need of Classification

Classification is essential to achieve following needs.

- To relate things based on common characteristic features.
- To define organisms based on the salient features.
- Helps in knowing the relationship amongst different groups of organisms.
- It helps in understanding the evolutionary relationship between organisms.

1.3.2 Classification of Living World

A comparison of classification proposed for classification of living world is given in Table 1.4.

1.3.3 Five Kingdom Classification

R.H.Whittaker, an American taxonomist proposed five Kingdom classification in the year 1969. The Kingdoms include **Monera,**

Table 1.4: Systems of Classification

Two Kingdom	Three Kingdom	Four Kingdom	Five Kingdom
 **Carl Linnaeus (1735)**	 **Ernst Haeckel (1866)**	 **Copeland (1956)**	 **R.H. Whittaker (1969)**
1. Plantae 2. Animalia	1. Protista 2. Plantae 3. Animalia	1. Monera 2. Protista 3. Plantae 4. Animalia	1. Monera 2. Protista 3. Fungi 4. Plantae 5. Animalia

XIth_BIO-BOTANY_Ch_1-EM.indd 8 5/17/2020 11:09:28 PM